

Logic Tree Initialization

Semantic Chunking

INTRODUCTION

Detection and prediction of emerging tipping points are major challenges ...

BACKGROUND

Transfer entropy (TE) is a ...

EVALUATION

Figure 1 is outside the prediction interval, indicating the instability ...

CONCLUSION

We propose a novel information ...

Research Paper

Semantic Coherence

Detection and prediction of emerging tipping points are ...

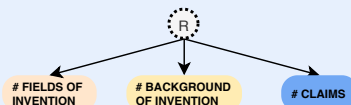
Contextual Dependency

Through the remainder of this paper, networked dynamical system ...

Consequently, network science has evolved from studying ...

Semantic Chunks

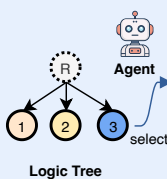
Tree Template



Logic Tree Growing

Context Retrieval

Point 3



A probabilistic light cone of instability trajectories is generated using natural logarithmic curve modeling, enabling analysis of system instability trajectories and providing early indications of critical transitions ...

Text	Sim	IsRet
C_1	0.72	T
C_2	0.44	F
C_3	0.35	T
...	...	...
C_N	0.24	F

Context Evidence

Text	Sim	IsRet
C_1	0.72	T
C_2	0.44	F
C_3	0.35	T
...	...	...
C_N	0.24	F

Context Reranking

Rerank
Relevance
Evidence
History

Text	Sim	IsRet
C_2	0.44	F
C_1	0.40	T
C_4	0.29	F
C_3	0.03	T

query retrieved

Context Retrieval History

update

select

Point 3

A probabilistic light cone of instability trajectories is generated using natural logarithmic curve modeling, enabling analysis of system instability trajectories and providing early indications of critical transitions ...

The process of generating the probabilistic light cone involves a series of steps, starting with the estimation of the coefficient ...

Point 3.1

The computation of the probabilistic light cone then proceeds with applying a moving time window ...

Point 3.2

The analysis of the behaviors of TE and ATE spectra includes examining the growth rate of the TE trajectory as information transfer approaches ...

Point 3.3

output

LLM

input

context

instruction

given a specific claim contribution, extract the concrete algorithmic steps from the provided context:

Text	Sim
C_2	0.44
C_1	0.40
C_4	0.29

Our proposed dynamic spectrum ...
Motivated by the ...
We develop novel ...

Prompt

Evidence-Guided Recursive Generation

format

M+

FIELDS OF INVENTION

The present invention relates to system for automated detection of critical transitions and ...

BACKGROUND OF INVENTION

One central characteristic of complex social-technological systems is their ...

...

CLAIMS

1. A method for analyzing system instability in complex systems, comprising:
- receiving multivariate time series data ...
- transforming the time series data ...
- calculating the spectral radius ...
- generating probabilistic light cones of ...
2. As described in claim 1, the method is characterized in that in step 1, transforming time series data into a transfer entropy matrix and ...
3. As described in claim 1 ...

Hybrid Traversal Strategy

Logic Tree Traversal



Research

LogicTree-RAG



Intellectual Property